

MATE: Mask-Guided Adaptive Object Template Estimation and Tactile-Confirmed Execution for Human-to-Robot Handover

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Abstract—Human-to-robot handover is challenging when transparent surfaces and hand occlusion produce incomplete or unreliable RGB-D geometry, while vision alone cannot confirm contact during execution. We present **MATE**, a Mask-guided Adaptive object template estimation framework with Tactile-confirmed Execution. To recover reliable object geometry under transparency and occlusion, **MATE** fits an instance-specific object template to dual-view partial point clouds and refines its scale using cross-view mask agreement. A magnet-based tactile fingertip regulates gripper closure and release. Experiments on CORSMAL containers and household objects achieve average success rates of 89.8% and 91.7%, respectively, demonstrating the complementary benefits of geometric adaptation and tactile sensing. Additional details are available at <https://vision3d-lab.github.io/mate/>.

I. INTRODUCTION

Human-to-robot handover involves a robot receiving an object held by a human giver [1]. Unlike in static pick-and-place, the hand-held object may move during the interaction, requiring the robot to select a stable grasp that avoids the hand and to establish secure contact with the object during execution [2], [3]. Reliable handover therefore requires both robust geometric perception and contact-aware execution.

However, the giver’s hand occludes the object before grasping, while the approaching gripper obscures the contact region during execution. Transparent object surfaces further produce incomplete and noisy RGB-D observations [4], [5], introducing uncertainty in object extent and grasp target estimation. Vision alone also cannot reliably verify fingertip-object contact. Prior work has addressed reactive hand-aware grasping [3] and hand-object reconstruction [6], while tactile handover control has largely relied on predefined grasps without addressing geometric uncertainty [7].

We present **MATE**, a **M**ask-guided **A**daptive object template estimation framework with **T**actile-confirmed **E**xecution for human-to-robot handover. Our contributions are summarized as follows: (1) mask-guided template adaptation that derives a geometry-consistent object shape from partial and unreliable RGB-D observations, and (2) tactile-confirmed execution that reduces contact uncertainty during grasping and placement. Experiments on filled containers, empty transparent containers, and household objects demonstrate the complementary benefits of both components.

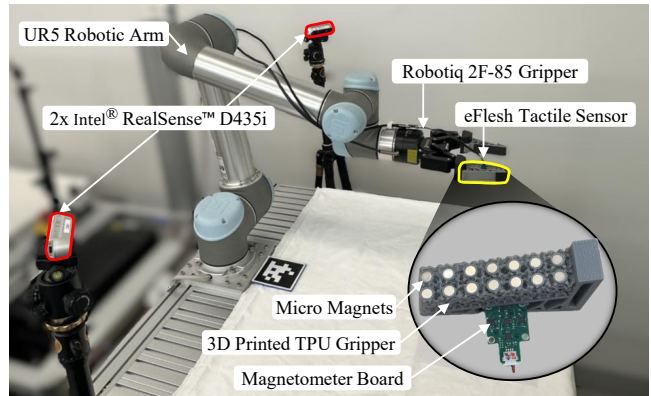


Fig. 1: Hardware configuration of MATE, comprising a UR5 robot, a Robotiq 2F-85 gripper with a magnet-based tactile fingertip, and dual RGB-D cameras.

II. PROPOSED SYSTEM

A. Hardware Configuration

As shown in Fig. 1, the dual-view configuration provides complementary observations of the hand-held object, reducing giver-hand occlusion and supporting object shape and grasp-target estimation. For close-range contact sensing, the custom TPU tactile fingertip uses an eFlesh [8]-style internal microstructure with micro-magnets positioned above a magnetometer array to measure contact-induced magnetic-field changes during grasping and placement. The workspace arrangement follows the human-to-robot handover setup of the CORSMAL benchmark [9].

B. Perception Pipeline

Object Template Fitting. The object-segmentation stage in Fig. 2 uses a hand-object detector [10] to localize the object in each RGB view and FastSAM [11] to extract the corresponding object mask. Valid masked depth pixels are lifted to 3D, filtered, and merged across the two views. During initialization for each new object, SAM3D [12] generates an instance-specific object template $\mathcal{T}_{gen}$ from a masked RGB observation. The initial scale is obtained by matching the oriented bounding box (OBB) extents of $\mathcal{T}_{gen}$ and the merged point cloud, followed by ICP registration of the scaled template to the point cloud. To compensate for missing or distorted depth, candidate templates with scales sampled around the initial estimate are projected into both camera views. The IoU between the projected template silhouette and the observed object mask is computed in each view, and the scale factor with the highest combined IoU across both views defines the refined template $\mathcal{T}_{ref}$.

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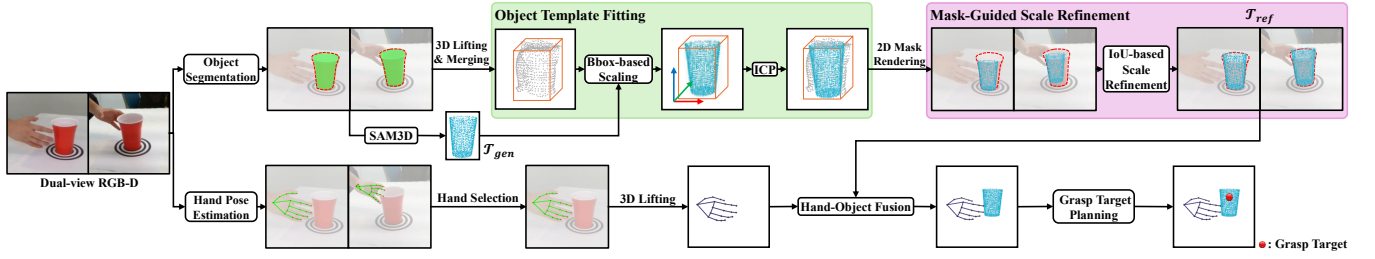


Fig. 2: Perception pipeline of MATE. Dual-view masks and depth form a merged partial point cloud, while a SAM3D-generated template $\mathcal{T}_{gen}$ is fitted using bounding box-based scaling and ICP and is then scale-refined by multi-view mask IoU to obtain $\mathcal{T}_{ref}$. Estimated hand landmarks by MediaPipe provide the 3D hand state, which is combined with $\mathcal{T}_{ref}$ for hand-aware grasp selection and online target updates.

Method	CORSMAL Containers							Household Objects				
	C1	C2	C3	C4	C3	C4	Avg.	O1	O2	O3	O4	Avg.
w/o Tactile	66.7	88.9	88.9	61.1	55.6	72.2	72.2	100.0	66.7	100.0	33.3	75.0
w/o Shape Fit.	94.4	66.7	100.0	72.2	50.0	55.6	73.2	50.0	33.3	66.7	0.0	37.5
w/o Scale Ref.	83.3	94.4	94.4	83.3	77.8	77.8	85.2	100.0	100.0	100.0	66.7	91.7
MATE (Ours)	88.9	94.4	100.0	83.3	83.3	88.9	89.8	100.0	83.3	100.0	83.3	91.7

TABLE I: Handover success rates (%) across evaluation objects and conditions. **C3** and **C4** denote empty transparent conditions; the evaluation objects are shown in Fig. 3.

Hand-Aware Grasp-Target Planning. 2D hand landmarks and handedness are estimated from both RGB images using MediaPipe [13]. The estimated handedness is used to select the view expected to be less occluded, after which the selected hand landmarks are lifted to 3D using depth images. The resulting hand state is fused with $\mathcal{T}_{ref}$. Among candidate points on $\mathcal{T}_{ref}$, a grasp target is selected to maintain clearance from the human hand while remaining close to the object center. The target is continuously updated in real time as the hand-held object moves. During temporary object-detection failures caused by occlusion, the target position is inferred from the tracked hand state using the previously estimated hand-to-target offset.

C. Tactile-confirmed Execution

While visual perception provides the grasp target, the magnet-based tactile fingertip controls close-range grasping and placement. Let n_t denote the norm of the baseline-subtracted magnetic response. After reaching the target, the gripper closes until n_t exceeds the grasp threshold τ_g , records the contact position, and performs a small additional closing motion to secure the object. Before placement, MATE records a reference value n_r and monitors $\Delta n_t = |n_t - n_r|$ while descending. When Δn_t exceeds the placement-contact threshold τ_p , the robot stops and releases the object.

III. EXPERIMENTS

Evaluation Protocol. Figure 3 shows four CORSMAL containers and four household objects spanning variations in geometry, transparency, and deformability. Success required stable upright placement; bean spillage counted as a failure. Each CORSMAL condition comprised 18 trials (2 participants $\times$ 3 grasp types $\times$ 3 locations), and each household object comprised six trials (2 participants $\times$ 3 locations).

Analysis. As shown in Table I, each ablation revealed a distinct failure mode in the handover pipeline. Without tactile

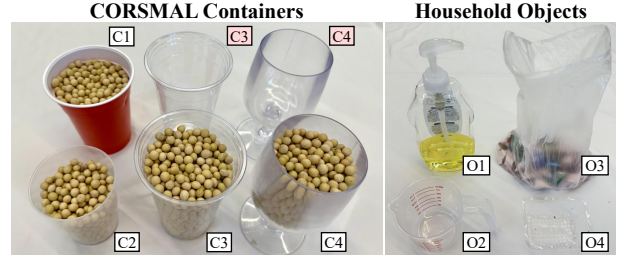


Fig. 3: Evaluation objects: CORSMAL containers C1–C4 (red cup, small cup, beer cup, and wine glass; C3–C4 were tested filled and empty) and household objects O1–O4 (soap, beaker, candy-filled plastic bag, and food container).

feedback, the absence of contact cues caused the gripper either to continue closing after object contact or to release the object before stable contact with the table, resulting in excessive closure and object drops, respectively. Without shape fitting, missing depth on transparent objects left the raw merged point cloud incomplete, sometimes preventing the identification of valid grasp targets and causing grasp failures. Without scale refinement, over-scaled templates caused grasp targets to fall outside the physical object boundary, leading to missed grasps. By addressing contact uncertainty, incomplete geometry, and scale errors at complementary stages, MATE maintained at least 83.3% success across all conditions, achieving the strongest worst-case performance.

IV. CONCLUSION

In this work, we presented MATE, a human-to-robot handover system that addresses geometric uncertainty and contact-state uncertainty during execution. Across diverse object types and depth conditions, MATE achieved at least 83.3% success rate, demonstrating robust grasping and placement under incomplete RGB-D observations.

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